

Cyberbullying Victimization and Mental Health Symptoms Among Children and Adolescents: A Meta-Analysis of Longitudinal Studies

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Abstract

Cyberbullying victimization and mental health symptoms are major concerns for children and adolescents worldwide. Despite the increasing number of longitudinal studies of cyberbullying and mental health among this demographic, the robustness of the causal associations between cyberbullying victimization and the magnitude of mental health symptoms remains unclear. This meta-analysis investigated the longitudinal impact of cyberbullying victimization on mental health symptoms among children and adolescents. A systematic search identified primary studies published in English between January 2010 and June 2021, yielding a sample of 27 studies encompassing 13,497 children and adolescents aged 8 to 19 years old. The longitudinal association between cyberbullying victimization and mental health symptoms among children and adolescents was found to be weakly positive and consistent across time and age. Three significant moderators were identified: the effect of cyberbullying victimization on mental health was larger among older children, groups with a higher proportion of males, and in more recent publications. No evidence of publication bias was detected. This study adds to the existing body of research by providing a new perspective on the long-term effects of cyberbullying victimization on the mental health of children and adolescents' mental health. Furthermore, it underscores the necessity of developing effective cyberbullying prevention programs, interventions, and legal regulations to comprehensively address this issue.

Keywords

longitudinal studies, cyberbullying victimization, mental health symptoms, children, adolescents, meta-analysis

Introduction

With the rise in digital adoption, cyberbullying victimization and its effects on mental health symptoms have become major concerns for children and adolescents worldwide. Cyberbullying refers to an act of aggression that occurs intentionally and is repeatedly carried out by an individual or a group using electronic media, such as chatrooms and instant messaging, to cause harm to a person who is unable to defend himself or herself (Smith et al., 2008). Lee et al. (2017) defined cyberbullying victimization as “being the object of aggressive or harmful behavior by others” via any types of electronic communications technologies (p. 457). Cyberbullying emerges on messaging, gaming and social media and smartphones, and the examples of cyberbullying behavior include sending hurtful or threatening messages, images or videos via messaging platforms; spreading gossip or rumors about or posting embarrassing images or videos of someone on social media; impersonating someone and sending mean messages to others on their behalf or through fake accounts; socially excluding someone in chat rooms (Lee et al., 2017; United Nations International Children's

Emergency Fund, 2020). According to a review of studies, about 14% to 58% of youths have experienced cyberbullying at some point in their lives (Zhu et al., 2021). Bullying victimization was found to be one of the leading global risk factors for mental health problems in the 2017 Global Burden of Disease study (Gakidou et al., 2017). Similarly, cyberbullying victimization is associated with a variety of mental health indicators, such as depression (Calvete et al., 2016; Lee, 2021; Schneider et al., 2012; Zhong et al., 2021), anxiety (Chu et al., 2019; Lee, 2021; Lee et al., 2021), low self-esteem (Lei et al., 2020; Palermi et al., 2017), and posttraumatic stress disorder (PTSD; Holfeld & Mishna, 2021).

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Effects of Cyberbullying Victimization on Mental Health Symptoms

Several theories explain how cyberbullying victimization may affect children and adolescents' mental health. For instance, according to the stress exposure model and vulnerability-stress model for depression, exposure to negative or stressful life events may increase the likelihood of developing mental health symptoms such as depression (Alba et al., 2018; Cole et al., 2006). Individuals' passive responses to stressors, particularly those arising from interpersonal relationship issues, are considered contributors to adverse mental health symptoms. Cyberbullying victimization is a significant stressor in the lives of children and adolescents, because these aggressive acts can be perceived as negative peer feedback and social rejection, reinforcing the victim's negative self-evaluations and potentially causing mental health problems (van den Eijnden et al., 2014). Furthermore, research reports that children and adolescents who have experienced cyberbullying victimization are less likely to disclose their incidents compared to those of traditional bullying (Dooley et al., 2010; Lee, 2021) due to fear of retaliation of cyberbullying, fear of loss of access to digital technology, and lack of confidence in parents or teachers' ability to help (Cassidy et al., 2013; Dennehy et al., 2020). Numerous studies indicate stronger relationships between cyberbullying and depressive and anxiety symptoms than traditional bullying (Bonanno & Hymel, 2013; Campbell et al., 2013; Cole et al., 2016; Tennant et al., 2015). As a result, victims of cyberbullying are more inclined to experience internalizing problems¹ such as depression and anxiety.

Previous empirical studies have shown a positive link between traditional bullying victimization and internalizing problems. Two decades ago, the first meta-analytic review of cross-sectional studies found strong positive correlations between traditional bullying victimization and psychosocial distress (Hawker & Boulton, 2000). Due to the increasing trends of cyberbullying victimization incidents among children and adolescents in contemporary society, several studies investigated the longitudinal associations between cyberbullying victimization and mental health problems, including depression, anxiety, loneliness, negative cognitions, somatic complaints, rumination, and PTSD (e.g., Chu et al., 2019; Fahy et al., 2016; Holfeld & Mishna, 2021; Jose & Vierling, 2018; Stapinski et al., 2015).

In the past, several meta-analyses reviewed bullying victimization and mental health. To date, several meta-analyses have examined the link between bullying victimization and mental health, but all had focused on traditional bullying instead of cyberbullying (e.g., Hawker & Boulton, 2000; Reijntjes et al., 2010), or had focused on multiple types of bullying (e.g., Christina et al., 2021; Eyuboglu et al., 2021; Li et al., 2024; Yuchang et al., 2019). For example, Reijntjes

et al.'s (2010) meta-analysis did not consider cyberbullying victimization specifically in its definition of peer victimization. Therefore, with the updated definition of cyberbullying, it is timely to conduct a more comprehensive review synthesizing the most up-to-date studies. Moreover, Christina et al. (2021) examined the bidirectional association between all types of bullying victimization (i.e., physical, verbal, relational, and cyberbullying) and internalizing problems among school-aged children. However, it mainly included studies about traditional bullying victimization, with only 13 studies related to cyberbullying victimization. It is important to focus on the effects of cyberbullying victimization on children and adolescents in particular, given the high degree of digital usage among children and adolescents, as well as greater daily stressors that they face which put them at a higher risk of developing mental health problems (Charles et al., 2013; Kim et al., 2019). The existing meta-analyses addressing the association between cyberbullying victimization and mental health included studies on cyberbullying victimization and depression (Hu et al., 2021; Tran et al., 2023), cyberbullying victimization and depression and anxiety (Yuchang et al., 2019), and the link between cyberbullying victimization and internalizing problems (Fisher et al., 2016; Kowalski et al., 2014). Only a few meta-analyses investigated the longitudinal association between traditional bullying victimization and internalizing problems (Reijntjes et al., 2010) and between differential bullying victimization and internalizing problems (Christina et al., 2021). Although these meta-analyses indicate that being cyberbullied is associated with negative mental health symptoms among children and adolescents, it remains unknown whether a longitudinal association holds across studies and if so, how strong this effect is. Given that most past meta-analyses had used cross-sectional data, we included only studies with longitudinal data so as to examine the temporal causal links between cyberbullying victimization on the one hand, and children's and youth's mental health outcomes on the other.

Potential Moderating Influences

There are mixed findings pertaining to the effects of potential moderators in the longitudinal associations between cyberbullying victimization and mental health symptoms. These include the gender and age of the victim, culture, publication year, and the time interval between when data on victimization and mental health outcomes were collected. Preliminary studies have shown that girls who have been bullied reported more internalizing symptoms than boys (Kowalski et al., 2014). Adolescent females were also more likely than males to report bullying victimization involving sexual behavior and activities (Brody & Vangelisti, 2017), and they experienced more emotional effects from the sexually explicit content involved in bullying victimization

(Sánchez et al., 2017). In addition, girls were more sensitive to online verbal bullying compared to boys, which could be attributed to how girls are often taught from a young age to be more attuned to the feelings of others (Bennett et al., 2005). On the other hand, Cénat et al. (2019) found boys reported higher prevalence of rumor spreading cybervictimization using a fake identity, and yet others reported no gender difference (Tennant et al., 2015). In view of the heterogeneity of findings, gender of the child was included as a moderator in the current meta-analysis.

Furthermore, previous research has found age differences in the link between cyberbullying victimization and mental health symptoms. Specifically, cyberbullying victimization had a nonlinear relationship with age; the percentage of victimization rose from 6.4% in 6th grade up to 11.6% in 10th grade, and subsequently fell to 7.8% in 12th grade (Lessne & Harmalkar, 2013). Recently, Pichel et al. (2021) found that cyberbullying victimization among adolescents aged 10 to 17 presented an inverted u-shaped relationship with a peak among the 14- to 15-year-old groups, and the specific items of cyberbullying victimization presented different prevalence rates across different age groups. For instance, 12- to 13-year-olds reported higher rates of rumors spread about them online, 14- to 15-year-olds presented higher rates of having nasty things said about them to others either online or through text messages, and 16- to 17-year-olds showed more experiences of having pictures or videos they posted online altered (Pichel et al., 2021). Moreover, Kowalski and Limber (2013) reported that middle school students who were victims of cyberbullying had higher rates of depression than high school students.

Cultural differences may also influence the link between cyberbullying victimization and mental health problems (Ferreira et al., 2016). A quantitative review found that cultures which value the importance of interdependence and harmony within the social group and high moral discipline had lower levels of aggression than other cultures where these values are less central to their identities (Bergeron & Schneider, 2005). For example, a 2-month longitudinal study revealed higher levels of cyberbullying among American students than Japanese students, which the authors attributed to a greater emphasis on interdependence and group harmony in Japan (Barlett et al., 2014). A meta-analysis by Yuchang et al. (2019) compared the effects of cybervictimization on depressive symptoms in children and adolescents across three cultures—North America, China, and Europe (including Australia)—and found that the effect size based on North American samples was higher than that of the Chinese and European samples. Although these cross-national differences may not be fully attributed to cross-cultural differences, especially given that Chinese and European samples were combined as one group in this study, there are still some valuable insights and interpretations. The authors attributed this finding to differences in self-construal, where

children and adolescents from Asian region are likely to attribute a bully's behavior to situational factors (e.g., the person having a bad day) rather than the individual's characteristics. This may account for why Asian children and adolescents may be more protected from the negative effects of cyberbullying. Overall, the results from existing research suggest that there may be differences in the experiences of cyberbullying among children and adolescents in Asian and Western cultures. However, most studies were conducted in the Western contexts, and comparative data from Asian societies is lacking (Li, 2008). A meta-analysis could provide a suitable methodology for synthesizing empirical studies from various countries and investigating the moderating effects of study locations and cultural contexts.

Publication year is also a potential moderator. As technology evolves, the prevalence of cyberbullying may increase (Slonje & Smith, 2008). A recent meta-analysis also found that publication year was a significant moderator in the relationship between cyberbullying victimization and depression, with a larger effect size observed in more recent publications (Hu et al., 2021). Furthermore, due to the unavoidable social isolation, work-from-home arrangements, and online learning brought about by the COVID-19 pandemic, the cyberspace has become a breeding ground for bullying to take place, with an adverse effect on children and adolescents' mental health symptoms since 2019 (Yang, 2021). Thus, the link between cyberbullying victimization and mental health symptoms might have become stronger between 2019 and 2021 compared to the other periods.

Time interval, which is the time elapsed between the time when cyberbullying victimization was first measured to the last time point when data on mental health symptoms were collected, may also be another potential moderator that has not been considered in previous meta-analyses. It is possible that the connection between bullying victimization at an early point and depression in longitudinal studies may have faded as other factors can also contribute toward depression (Winding et al., 2020). Children and adolescents may experience life events other than bullying victimization (e.g., being in a romantic relationship) over a period of several years, and these events can buffer (or exacerbate) the effects of cyberbullying victimization on mental health symptoms. As a result, the effect of cyberbullying victimization on negative mental health outcomes may fade over time if protective factors are present within that time interval (Holfeld & Mishna, 2021).

The Present Study

The present study aimed to conduct a meta-analysis of the existing research literature to investigate the longitudinal impact of cyberbullying victimization on mental health symptoms and identify factors that may moderate this link. Identifying potential moderators is a particularly important contribution because this relationship is likely to differ

across age, the child's gender, culture, publication year, and time interval. To this end, only longitudinal studies that examined cyberbullying victimization and various types of mental health symptoms that followed the same group of children and adolescents for at least two different time points were considered. Although there have been several systematic reviews and meta-analyses on the association between cyberbullying and mental health, to our knowledge, few meta-analyses have exclusively addressed the long-term effect of cyberbullying victimization on mental health symptoms among children and adolescents. Given the severity and pervasiveness of cyberbullying victimization for younger populations and its long-lasting negative consequences, a meta-analysis of the studies on the long-term effects of cyberbullying victimization on mental health symptoms among children and adolescents is timely to inform future research directions, as well as practice and policy implications.

Hypotheses

In light of these considerations, the following hypotheses were developed. Foremost, a positive relationship between cyberbullying victimization and poorer mental health symptoms is expected over time, specifically depression, anxiety, and psychological distress. This hypothesis was developed based on the results of cross-sectional studies, longitudinal studies, and previous meta-analyses that supported the link between cyberbullying victimization and its negative effects on mental health (Chu et al., 2019; Fisher et al., 2016; Kowalski et al., 2014; Yuchang et al., 2019). No specific hypothesis was formulated on the effects of the potential moderators given the mixed findings in the existing literature, and this remains an exploratory analysis. However, specific hypotheses were proposed for age, the child's gender, and year of publication, based on past research.

Method

Search Strategy and Screening of Studies

A literature search was conducted by two authors (JL and YZ) on eight electronic databases (PsycINFO, SCOPUS, Web of Science, PubMed, MEDLINE, ERIC, ProQuest, and Google Scholar) for studies published in English between January 2010 and June 2021. Additional searches were conducted from the reference lists of collected publications. Search strings can be found in Supplemental Appendix A. We followed the PRISMA guidelines when conducting the meta-analysis (Shamseer et al., 2015).

Studies were included if they met the following inclusion criteria: (a) written in English, (b) quantitative longitudinal design with at least two waves of data, (c) a study population of children² and adolescents³ aged 8 to 19 years, (d) included cyberbullying victimization as an independent variable, and

(e) included mental health indicators (i.e., depression, anxiety, psychological distress) as a dependent variable. Publications were excluded if they sampled young adults, university (college) students, emerging adults, or any older population groups; referred to cyberbullying perpetration only, included traditional forms of bullying or cybercrimes (e.g., cyber scam, cyber pornography, cyber stalking, online sexual or dating violence), or if the data were derived from qualitative studies, case reports, opinion papers, intervention, or experimental studies. To be included in the synthesis process (meta-analysis), the full-text article of any study had to meet the following criteria: (a) reported either correlation coefficient (β) or r , t , and F values that could be transformed into β values; and (b) measured quantitative data from at least two waves. Studies were excluded when there were duplicate samples.

After the initial search of articles, a stepwise screening process was performed by three independent authors (JL, YZ, and QZ) as shown in Figure 1. The first step involved the removal of duplicates. Next, the title and abstract of the remaining articles were screened for relevance to the study. Studies that met the inclusion criteria based on title and abstract were then brought forward to the next stage of full-text articles screening to be assessed for eligibility. Two authors independently reviewed the full text for the studies to assess the study eligibility and discussed seeking for inter-rater agreement. Then, the first author resolved any discrepancies in the research team. The PRISMA chart appears in Figure 1. We followed the practice recommended by Polanin et al. (2019) to arrange regular team meetings to reconcile disagreement and reduce interpretation inconsistencies. Although there were some conflicts of decisions emerging from time to time, we resolved these conflicts and reached a consensus with 100% inter-rater agreement. After full-text review, 27 studies were included in the meta-analysis.

Data Extraction and Coding in the Meta-Analysis

For 27 longitudinal studies that met the inclusion criteria through full-text screening, three authors (JL, YZ, and QZ) were involved in the data extraction and coding process. Similar to the screening process, we held bi-weekly meetings to resolve discrepancies in coding. Each study was coded independently by two authors as per the recommendation by Bergstrom and Taylor (2006), and discrepancies were discussed to agreement. We coded the following information from each paper in the meta-analysis: longitudinal study design (i.e., names of authors, year of publication, culture, number of waves, time interval—in months—between the first and the last wave); study sample (i.e., sample size, percentage of females, mean age of participants in the first wave); cyberbullying victimization measures as a predictor; and mental health symptoms as an outcome. The countries of all the studies included the

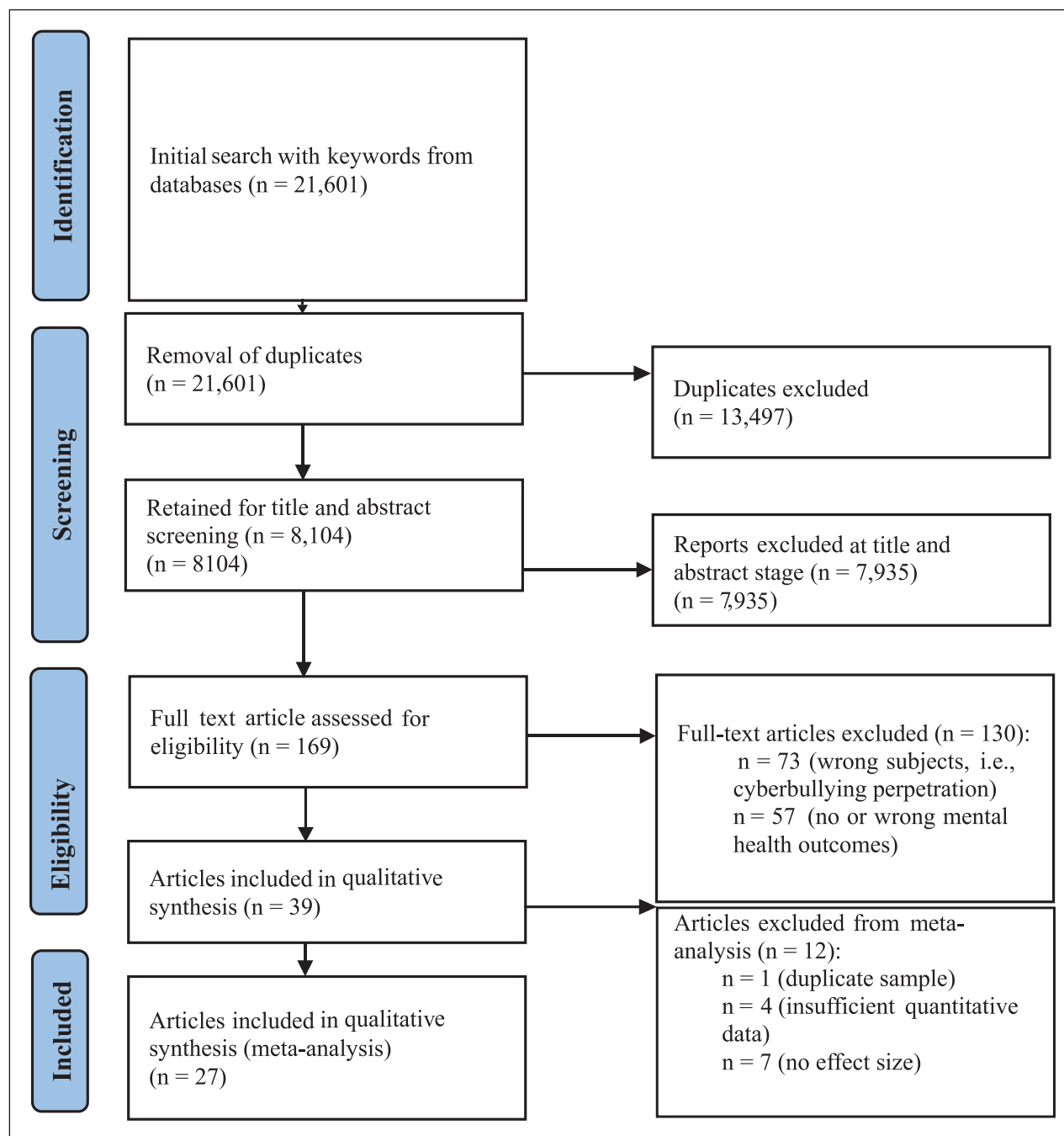


Figure 1. PRISMA flow diagram showing process of study selection for inclusion in meta-analysis.

United States ($n=12$), China ($n=7$), Spain ($n=2$), Sweden ($n=1$), Netherlands ($n=1$), Belgium ($n=1$), Switzerland ($n=1$), Australia ($n=1$), and New Zealand ($n=1$). In the meta-analysis, we recoded these countries as one variable (culture with two categories, Western and Asian culture). As a measure of effect size, correlation coefficients were coded. Correlation coefficients were averaged across studies that reported multiple prospective waves. For example,

if three correlation coefficients between cyberbullying victimization and mental health symptoms were reported for each T1 to T2, T1 to T3, and T2 to T3, an average of the three values was used for analysis. If several mental health symptoms were reported in the same study, the averaged correlation coefficients of all the different mental health symptoms were used. Thus, each study produced only one correlation coefficient for analysis.

Methodological Quality Assessment

For the assessment of methodological quality of each article, five methodological study characteristics were considered: (a) sampling procedure (probability sampling vs. nonprobability sampling); (b) dropout rate (in %) between the first and the final wave; (c) systematic dropout check between the initial and the analytical sample; and (d) type of missing data handling (see Supplemental Appendix C).

Statistical Analyses

A random-effects model was used to analyze the data since there is a distribution of true effect sizes instead of all studies measuring the same true effect size (Borenstein et al., 2010). In addition, the homogeneity test for cyberbullying victimization and mental health symptoms showed significant heterogeneity ($Q=335.48$, $p<.001$; $I^2=92.25\%$), which required a random-effects model to calculate the pooled correlation coefficient. We ran moderator analyses between included effect sizes based on mean age, the child's gender (the percentage of females in the sample), culture, and year of publication. Previous studies have demonstrated that these variables play significant moderating roles in the association between bullying victimization and mental health (Marciano et al., 2020; Yang, 2021). Subgroup analysis was used to determine whether culture is a moderator. Finally, we used meta-regression to determine whether female percentage, mean age, time interval, and year of publication were significant moderators. Funnel plot, Egger's regression test, and Rosenberg's Fail-Safe N were used to assess the possibility that publication bias influenced our results. All these analyses were carried out using software comprehensive meta-analysis.

Results

Study Characteristics

Table 1 lists the 27 included studies and their characteristics. The sample sizes of the included studies ranged from 113 to 3,961. Studies were conducted across nine countries, with most of them taking place in the United States (44.4%) and China (25.9%). The total sample size from all the studies was 27,133 participants. Fourteen of the 27 studies (59%) reported two-wave longitudinal studies, and 13 were three-wave longitudinal studies. All the studies found a positive link between cyberbullying victimization and mental health symptom indicators. The studies also included a range of mental health symptoms: 21 studies focused on depression; 16 on anxiety; 3 on loneliness; 3 on negative cognitions, low self-esteem, and body image or psychological distress; 3 on somatic complaints, sleep problems or stress; 2 on posttraumatic stress symptom/PTSD; and 1 on rumination, mental well-being, or life satisfaction.

Longitudinal Effects of Cyberbullying Victimization on Mental Health Symptoms

A significant positive correlation between cyberbullying victimization and ill effects on children and adolescents' mental health over time was found in the meta-analysis ($r=.23$, $p<.001$; 95% CI [0.19, 0.27]; $n=27$). Figure 2 shows a forest plot that illustrates the longitudinal relationship between cyberbullying victimization and overall mental health symptoms over time, along with effect sizes and 95% confidence interval (95% CI) for each study that was included in the meta-analysis. The five forest plots that show the longitudinal effects of cyberbullying victimization on (1) depression, (2) anxiety, (3) loneliness, (4) body image/negative cognition/low self-esteem/psychological distress, and (5) somatic complaints/sleep/stress respectively are additionally reported in Supplemental Appendixes D to H (Supplemental Material). In Table 2, we presented subgroup analyses on the individual outcomes. There was a significant and positive correlation between cyberbullying victimization and depression ($r=.27$, $p<.001$; 95% CI [0.21, 0.33]; $n=23$), and between cyberbullying victimization and anxiety ($r=.23$, $p<.001$; 95% CI [0.16, 0.29]; $n=16$). However, there were no significant associations between cyberbullying victimization and the other mental health outcomes, including loneliness, body image/negative cognition/low self-esteem/psychological distress, and somatic complaints/sleep/stress due to the small sample sizes.

Moderation Effects

We conducted subgroup analyses on dichotomous moderator (i.e., culture) and meta-regression on continuous moderators (i.e., age, female percentage, publication year, and time interval). Culture was not a significant moderator in the longitudinal association between cyberbullying victimization and mental health symptoms ($p=.730$). The average r is .25 ($n=7$) among Asian studies and .23 ($n=20$) among Western studies.

Table 3 presented the detailed statistics of the meta-regression analysis with moderators. The results showed a significant difference across age ($\beta=.04$, $p<.01$; see Supplemental Appendix I), female percentage ($\beta=-.003$, $p<.01$; see Supplemental Appendix J), and publication year ($\beta=.01$, $p<.001$; see Supplemental Appendix K), but there was no moderating effect across time interval ($\beta=-.001$, $p=.526$; see Supplemental Appendix L). These results show that the longitudinal impact of cyberbullying victimization on mental health is larger among older children, children groups consisting of more males, and among more recent publications.

When we restricted the mental health outcomes to depression, age ($\beta=.05$, $p<.001$) and female percentage ($\beta=-.00$, $p<.01$) remained as statistically significant moderators. However, publication year was no longer significant ($\beta=.01$, $p=.07$), while time interval became significant ($\beta=.02$, $p<.001$). When we restricted the mental health outcomes to

Table 1. Study Characteristics for Articles Included in Meta-Analysis.

Authors (Year)	Year of Publication	Culture	Total Sample Size (N)	Female (%)	Mean Age	Time Interval (Month)	Cyber-Victimization Measurement (Item No.)	Mental Health Symptoms	Correlation Coefficient
Calvete et al. (2016)	2016-2018	Western	1050	57.9	15.43	12	CBQ (9)	Depression, body image	.117
Chu et al. (2019)	2019-2021	Asian	661	39.2	12.86	12	Chinese version of the Revised CBI (14)	Depression, anxiety, stress, loneliness	.205
Cole et al. (2016)	2016-2018	Western	827	55.1	10.90	6	PVSR (4)	Depression, negative self-cognition	.120
Diaz and Fite (2019)	2019-2021	Western	260	47.0	12.24	6	ECIPQ (6)	Depression, anxiety	.190
Gámez-Guadix et al. (2013)	2012-2015	Western	845	58.9	15.22	6	CBQ (9)	Depression	.256
Gao et al. (2021)	2019-2021	Asian	2104	49.9	12.75	12	Self-report CV (18)	Depression	.210
Herge et al. (2016)	2016-2018	Western	1162	57.0	15.80	4.1	C-PEQ (8)	Depression, anxiety, somatic complaints	.247
Hoffeld and Mishra (2021)	2019-2021	Western	510	61.6	13.70	6	Self-report CV (1)	PTSD	.140
Jose and Vierling (2018)	2016-2018	Western	2179	52.1	NA	12	Self-report CV (2)	Rumination	.163
Liu et al. (2020)	2019-2021	Asian	661	41.9	13.51	8	Chinese version of RCBPI-CS (14)	PTSD, depression	.280
Liu et al. (2021)	2019-2021	Asian	1274	46.2	15.19	6	Chinese version of RCBPI-CS (14)	Depression, anxiety	.489
Machmutow et al. (2012)	2012-2015	Western	765	52.1	13.18	6	Self-report (6)	Depression	.190
Pabian and VandeBosch (2016)	2016-2018	Western	2128	56.6	13.02	6	Self-report CV (1)	Anxiety	.090
Rose and Tynes (2015)	2012-2015	Western	559	55.1	NA	12	Online victimization scale (6)	Depression	.193
Tian et al. (2018)	2016-2018	Asian	606	53.3	NA	12	EBQ (4)	Stress, depression, anxiety	.137
van den Eijnden et al. (2014)	2012-2015	Western	836	50.2	13.20	24	Self-report CV (7)	Loneliness, anxiety	.085
Van Zalk and Van Zalk (2019)	2019-2021	Western	501	51.9	13.96	8	Self-report CV (9)	Depression, anxiety	.167
Wright et al. (2018)	2016-2018	Western	1090	50.0	12.19	12	Self-report CV (6)	Depression, anxiety	.315
Wright and Wachs (2019)	2019-2021	Western	416	46.0	13.89	12	Self-report CV (9)	Depression, loneliness	.257
Wright (2015)	2012-2015	Western	423	49.9	16.67	12	Self-report CV (13)	Depression, anxiety	.485
Wright (2016)	2016-2018	Western	867	51.0	13.67	12	Self-report CV (9)	Depression, anxiety	.337
Wright (2017)	2016-2018	Western	131	27.0	NA	12	Self-report CV (9)	Depression	.320
Wright (2018a)	2016-2018	Western	113	14.0	NA	12	Self-report CV (9)	Depression	.200
Wright (2018b)	2016-2018	Western	567	52.0	13.48	12	Self-report CV (9)	Depression, anxiety	.305
Zhang et al. (2020)	2019-2021	Asian	3961	43.9	10.85	6	Chinese version of the brief adaptation EBQ (4)	Negative cognition, depression	.137
Zhu et al. (2021)	2019-2021	Asian	1987	43.9	12.32	6	Chinese version of the brief adaptation EBQ (4)	Anxiety	.230
Zimmer-Gembeck et al. (2021)	2019-2021	Western	650	59.0	17.30	12	POTS (2)	Anxiety	.360

Note. CBQ = cyberbullying questionnaire, C-PEQ = Cyber-Peer Experiences Questionnaire, PVSR = peer victimization self-report, EBQ = Electronic Bullying Questionnaire, ECIPQ = European Cyberbullying Intervention Project Questionnaire, POTS = Perception of Teasing Scale, RCBPI-CS = Revised Cyber Bullying Inventory-Cyberbullying Subscale; PTSD = posttraumatic stress disorder.

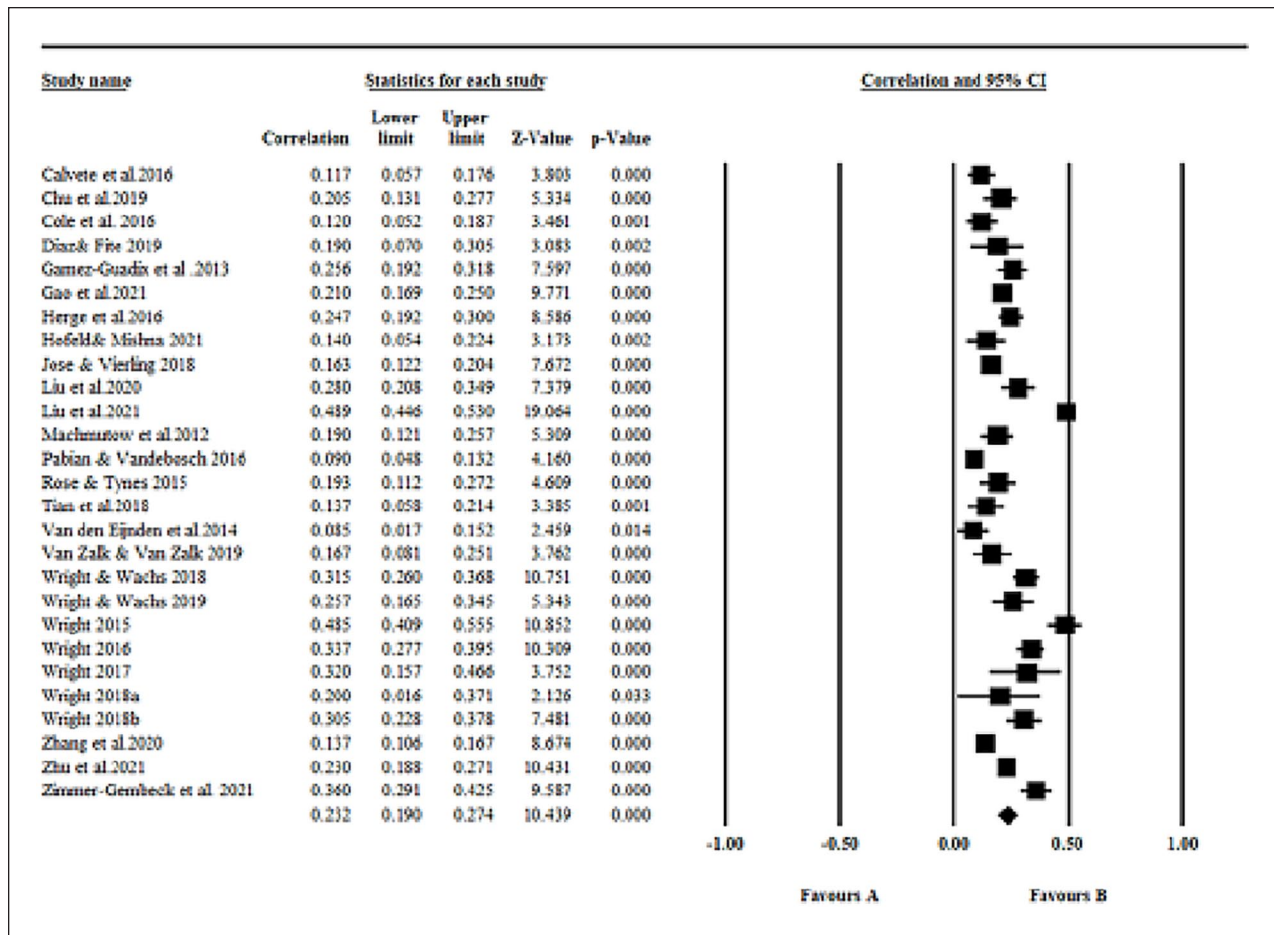


Figure 2. Forest plot of the longitudinal effects of cyberbullying victimization on mental health symptoms.

Table 2. Random-Effect Models of Correlations Between Cyberbullying Victimization and Each Mental Health Outcome.

Outcome	k	N	Mean r	95% CI for r [LL, UL]	Homogeneity Test			Tau-Squared			Test of Null Z-Value
					Q(r)	p	I ²	Tau-Squared	SE	Tau	
Overall mental health	27	27,133	.232	[0.190, 0.274]	335.477	.000	92.250	0.012	0.005	0.110	10.439
Depression	23	21,481	.269	[0.211, 0.325]	432.175	.000	94.909	0.021	0.008	0.144	8.799
Anxiety	16	13,987	.229	[0.162, 0.294]	253.119	.000	94.074	0.019	0.008	0.137	6.559
Loneliness	4	2,780	.170	[0.109, 0.229]	7.895	.048	62.003	0.002	0.003	0.049	5.465
Body image	3	5,838	.024	[-0.173, 0.219]	87.305	.815	97.709	0.030	0.033	0.173	0.235
Somatic complaints	3	5,784	.233	[0.202, 0.264]	2.580	.000	22.481	0.000	0.001	0.015	14.390

Note. CI = confidence interval.

Table 3. Meta-Regression Result.

Moderators	β	SE	95% CI [lower, upper]	Q_{between}	p
Age	.037	0.004	[0.030, 0.045]	94.329	.000
Female percentage	-.003	0.001	[-0.005, -0.001]	7.590	.006
Publication year	.009	0.002	[0.004, 0.013]	13.362	.000
Time interval	-.001	0.001	[-0.004, 0.002]	0.402	.526

Note. SE = standard error; CI = confidence interval; df = degree of freedom.

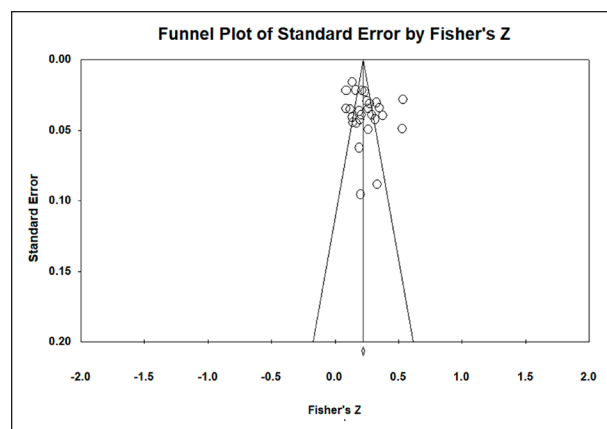


Figure 3. Result of funnel plot publication bias analysis.

anxiety, age ($\beta = .05$, $p < .001$), female percentage ($\beta = -.01$, $p < .001$), and publication year ($\beta = .02$, $p < .001$) remained statistically significant. Time interval also became statistically significant ($\beta = -.00$, $p < .05$).

Publication Bias

The funnel plot (see Figure 3) showed that data were generally distributed symmetrically around the combined effect size, and Egger's tests ruled out publication bias as a possible confounder ($t = 1.52$, $p = .071$). Rosenberg's Fail-Safe N revealed that 8,588 missing studies with effect sizes of zero were required to lower the p value to a statistically nonsignificant level, indicating there is little likelihood that the current findings are affected by publication bias.

Discussion

To understand the longitudinal link between cyberbullying victimization and mental health symptoms, data from 27 empirical studies involving 27,133 participants were examined using meta-analysis. The moderating effects of age, the child's gender, culture, year of publication, and time interval on the longitudinal link between cyberbullying victimization and mental health symptoms were also investigated. A summary of key findings of the review is presented in Table 4.

Longitudinal Effects of Cyberbullying Victimization on Mental Health Symptoms

Consistent with our hypothesis, we found that the overall effect size between cyberbullying victimization and mental health problems over time was significant, positive and moderate (Cohen, 1988). This finding is in line with the previous cross-sectional meta-analyses (Fisher et al., 2016; Hu et al., 2021; Kowalski et al., 2014; Yuchang et al., 2019), but this study provides more compelling and robust evidence by including only longitudinal studies. Cross-sectional studies

provide correlational evidence, while longitudinal studies suggest the direction of the relationship. This meta-analysis synthesizing longitudinal studies showed that there was a positive relationship from cyberbullying victimization to mental health symptoms among children and adolescents.

Moderating Effects of Age, Female Percentage, and Year of Publication

Consistent with our hypothesis, the impact of cyberbullying victimization on mental health symptoms increased with age. However, in contrast to our hypothesis, as the percentage of female participants in the sample rose, the impact of cyberbullying victimization on mental health symptoms decreased. The effect of cyberbullying victimization on mental health symptoms did not significantly differ by culture and time interval between assessments of cyberbullying victimization and mental health outcomes. These findings are discussed in turn.

First, we found that the effect size between cyberbullying victimization and mental health symptoms increased as the participants' mean age increased (age:8–19). This is consistent with previous meta-analysis's findings examining the moderating effect of age on the association between cyberbullying victimization and depression (Hu et al., 2021; Yuchang et al., 2019). One explanation was that older adolescents have more access to the internet and social media platforms than younger children, which may expose them to a greater risk of cyberbullying (Lenhart, 2010). Another possible explanation is that older adolescents are facing more complicated events and daily stressors than the younger ones and may become more susceptible to the negative effects of cyberbullying victimization. When they experience cyberbullying victimization coupled with daily stresses, their risk of internalizing problems increases (Charles et al., 2013).

Contrary to our hypothesis and the findings of previous studies, our study found that the effect size decreased as the female percentage increased (Hu et al., 2021; Yuchang et al., 2019). One possible reason is that males are more likely to become cyberbullying victims than females (Cénat et al., 2019; Festl & Quandt, 2016). Higher levels of cyberbullying victimization were found among males who were previously exposed to more antisocial online content (Festl & Quandt, 2016). When boys engage in antisocial online behaviors, they may interact more frequently with (online) groups where antisocial norms and behaviors are more widely accepted, increasing their risk of victimization. Accordingly, the magnitude of the link between cyberbullying victimization and mental health symptoms becomes weaker when female percentage is higher.

Researchers have disputed if the prevalence of cyberbullying is increasing or has leveled off over time (Kowalski et al., 2014). Some cross-sectional studies show that the prevalence of cyberbullying is rising as technology advances (Slonje & Smith, 2008; c.f. Olweus, 2013). In addition, a meta-regression of studies in the United States found a

Table 4. Summary of Critical Findings.

Findings	
• The longitudinal effect of cyberbullying victimization on mental health symptoms among children and adolescents was found in the meta-analysis. • The five forest plots showed the longitudinal effects of cyberbullying victimization on depression, anxiety, loneliness, body image/negative cognition/low self-esteem/psychological distress, and somatic complaints/sleep deprivation/stress, respectively. • There were three significant moderators: female percentage, age, and publication year. • The magnitude of the association between cyberbullying victimization and mental health symptoms became larger when children and adolescents were older, the female percentage was lower, and the studies were more recently published. • Cultural differences and time interval between assessments did not moderate the effect of cyberbullying victimization on mental health symptoms.	

Table 5. Implications for Research, Practice, and Policy.

Area	Implications
Research	• Further research is needed to produce more accurate results and test a wider range of moderators. • Further research would benefit from inclusion of diverse cultural contexts when more relevant research is available. • A more nuanced and holistic explanation of different types of cyberbullying is needed and can aid in the design of more effective intervention programs to tackle cyberbullying.
Practice	• Developing a positive parent-child emotional bonding is crucial to protect cyberbullying victimization and mental health symptoms. • For a greater need for autonomy among children and adolescents, the use of active mediation strategies instead of restrictive mediation may be more effective in protecting them from online harm.
Policy	• To mitigate potential risks in cyberspace, appropriate policies and laws should be created in the countries which lack online safety regulations. • Industry self-regulation should be encouraged and content providers should be socially responsible for the content generated. • As cyberbullying victimization has no borders and its impact on mental health symptoms remains universal and consistent across cultures, it would be advantageous to foster collaboration among regions of the world.

consistent increase in cyberbullying victimization from 2010 to 2017 (Kennedy, 2021). However, many included studies used the same sample, and only one was longitudinal, indicating a need for further investigation. Another cross-sectional study among teenagers in Australia concluded that cyberbullying prevalence did not change from 2015 to 2020, while cyberbullying victimization increased (Trompeter et al., 2022). The current meta-analysis included empirical cyberbullying studies published between 2012 and 2021 and found that the effect size for cyberbullying victimization and mental health symptoms differed across publication year, which is consistent with previous meta-analysis’s findings showing that the effect size for cyberbullying victimization and depression increased in more recent publications (Yuchang et al., 2019).

Moderating Effects of Culture and Time Interval

Consistent with previous findings (Hu et al., 2021), this study discovered that cultural differences did not moderate the effect of cyberbullying victimization on mental health symptoms. According to some reports, cyberbullying has become a global issue that transcends relevant circumstances and culture (Kraft, 2006). In other words, the Internet connects young people globally, and cyberbullying has become

a widespread problem that spans across diverse cultures (Lee et al., 2017). It is possible that the Internet may have an effect of flattening cross-cultural differences since it is itself a culture with its own norms; it subsumes country, race, ethnicity, and other subgroup differences and allows individuals to adopt, customize, and choose how they wish to present themselves online.

Furthermore, this study found that the effect of cyberbullying victimization on mental health symptoms did not significantly differ by time interval between assessments, which are somewhat surprising in light of Cole and Maxwell’s (2003) findings. Cole and Maxwell (2003) found that the relationship between time interval and effect size is curvilinear that is modeled by an inverted-U curve, where the effect size would become greater with a greater time interval, before peaking at a specific time interval and then drop to approach zero over time. One possible explanation could be that the number of studies was insufficient ($n=27$) for this type of moderator analysis to have enough effect sizes in each category, limiting statistical power.

Implications for Research, Practice, and Policy

Several implications related to research, policy, and practice arise from the study findings (see Table 5). Research on

cyberbullying victimization has grown in recent years, but there are very few quantitative syntheses of these studies. Only five related meta-analysis studies were found that have investigated the relationship between cyberbullying victimization and various mental health problems (Fisher et al., 2016; Hu et al., 2021; Kowalski et al., 2014; Tran et al., 2023; Yuchang et al., 2019). Previous meta-analyses only included studies that reported cross-sectional data, and this study extends current literature by meta-analyzing longitudinal studies that generated new insights and perspectives on the long-term effect of cyberbullying victimization on children and adolescents' mental health. Even though diverse mental health symptoms were examined in relation to cyberbullying victimization across at least two waves of data in 27 different studies, the association between cyberbullying victimization and the ill effects on mental health remained remarkably consistent across symptoms. This implies that the effect of cyberbullying victimization and mental health symptoms is noteworthy and very likely not spurious. These findings culminate in a call to action for researchers, policymakers, and practitioners to jointly consider prevention and intervention efforts to bolster the mental well-being of individual children and adolescents as well as the collective well-being of society and its various ecosystems.

When considering upstream prevention and intervention efforts to support children and adolescents who are victims of cyberbullying, it is important to leverage on the support figures in the victims' social networks, which include parents and peers. For children and adolescents, parents remain an influential and reliable source of support, even as peers play an increasingly important role starting from early adolescence. Developing a positive parent-child emotional bonding is therefore crucial as it serves as a protective factor from cyberbullying victimization. Parental mediation strategies are vital in helping children and adolescents recognize the risks and consequences of personal information disclosure along with learning ways of protecting their personal information and making discerning choices about the kinds of information they can reveal (Liu et al., 2013). For adolescents in particular, the use of active mediation strategies (e.g., having discussions about the potential dangers in cyberspace) may be more effective in protecting them from harm than restrictive mediation (e.g., limiting Internet use through rules and/or coercion), in view of their greater need for autonomy compared to younger children. The effectiveness of mediation strategies is in part dependent on the relationship children and adolescents have with their parents (Huang et al., 2024), and whether this relationship is emotionally strong, positive, and trusting.

Another contribution of study is by examining the moderating effect of time interval and re-examining the moderating effects of age, female percentage, culture, and publication year with different longitudinal mixed sample data. The efficiency of legal regulations against cyberbullying in some regions is unknown (Foody et al., 2017), and many countries

(e.g., China) continue to lack such regulations. As a result, there is an urgent need to create appropriate policies and laws to mitigate potential risks in cyberspace. For these country-level regulations and guidelines to be effective, they need to take into account the context and the needs of the specific populations. In addition to regulations and laws, it is equally vital to encourage industry self-regulation, and for content providers to be socially responsible for the content generated. Furthermore, because cyberbullying victimization has no borders and its effect on mental health symptoms remains common across cultures, it would be advantageous to foster collaboration among regions of the world to establish shared understanding about this phenomenon, and to develop some international standards and guidelines.

Limitations and Future Research

There are several limitations to the current review. First, only 27 independent studies were included in the meta-analysis. As more longitudinal studies become available, future meta-analyses may add validate these findings and potentially test a wider range of moderators. The small sample size also limits the power and generalizability of our meta-regression results; therefore, readers should interpret the meta-regression analysis with caution. Second, this meta-analysis focused on only searchable literature published in English. Future studies can include research published in non-English languages which may potentially expand the selection of studies from diverse cultures. Third, the majority of the studies in our meta-analysis had focused on depression and anxiety, with only a few studies on other mental health symptoms. Therefore, conclusions about other mental health symptoms than depression and anxiety should be interpreted with caution. Fourth, the result of culture not being a moderator should also be treated with caution, as we only compared Asian and Western cultures in the moderating effect of culture on the longitudinal link between cyberbullying victimization and mental health symptoms over time. Future meta-analysis would benefit from inclusion of other cultural contexts when more related research is available. Finally, this study did not investigate the diversity in experiences of cyberbullying victimization, such as types of the technological platform (e.g., phone conversations, texting, e-mails, pictures, videos). Because technological advancement leads to varied platforms for cyberbullying, a more nuanced explanation and comprehension of the experiences of different kinds of online bullying is needed and can aid in the design of more effective cyberbullying interventions (Khan et al., 2020). Primary studies should consider these factors when investigating this association.

Conclusion

This meta-analysis research aimed to determine the longitudinal effect of cyberbullying victimization on mental

health symptoms among children and adolescents and examine moderators for the effect. The meta-analysis of 27 longitudinal studies published in January 2010 to June 2021 revealed that the negative effect of cyberbullying victimization on mental health among children and adolescents was notably consistent across various mental health symptoms. Furthermore, with the exception of age, female percentage, and year of publication, the absence of a significant moderating effects of culture and the time elapsed between the onset of victimization and the measurement of mental health outcomes suggests that the detrimental impact of the victimization experience on mental health persists across different spaces, time periods and research contexts. Given the evident mental health consequences of cyberbullying victimization among children and adolescents, the importance of preventive programs for cyberbullying as well as professional interventions to address its effects on mental health should be emphasized. Future research—both primary studies and meta-analyses—will be needed to identify additional moderators and potential mediators in advancing our knowledge of how cyberbullying victimization may affect mental health among young people, and the intervening factors involved to better inform prevention and intervention strategies.

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Notes

1. Internalizing problems refer to inward experiences and symptoms such as sadness, anxiety, and loneliness. This is opposite of externalizing problems, such as aggression and hyperactivity (Levesque, 2011).
2. Defined as every human being below the age of eighteen years (UNESCO, 2020).
3. Adolescence is defined as the transitional phase between childhood and adulthood, which spans ages 10 to 19 (WHO, 2024).

Supplemental Material

Supplemental material for this article is available online.

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